



An AI-powered conversational system for college students learning English as a second language

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Abstract

With the advancement of artificial intelligence, natural language processing, and speech recognition, conversational agents have emerged as promising tools for second language acquisition. This study designed an English conversational bot to help learners of English as a second language among Chinese university students learning English. The bot was implemented as a web application using Python Flask. A six-day comparative study was conducted with 56 students from the English department, who were randomly assigned to either the conversational bot or a traditional listen-and-repeat interface. The bot was implemented as a web application using Python Flask as a web application. To evaluate its feasibility, we compared it with an old listen-and-repeat interface using 56 Chinese college students from the English department over six days. We conducted Shapiro-Wilk normality tests, followed by t-tests to evaluate time spent on the application, engagement, anxiety change, vocabulary, and speaking tests. Findings shows that the conversational bot group reported significantly higher engagement scores ($M=4.37$, $SD=0.44$) compared to the control group ($M=3.87$, $SD=0.51$; $t(26)=2.8$, $p<.05$). While both groups showed a reduction in language anxiety, the difference was not statistically significant. Participants using the bot demonstrated greater vocabulary use as responders ($t(26)=3.5$, $p<.005$) and higher speaking test gains ($p<.0001$). In conclusion our findings suggest that the conversational bot is a more engaging and effective platform for improving spoken English proficiency. The tool shows potential for supporting learners in preparing for international academic environments.

Keywords Artificial intelligence · Conversational bots · English learning · Natural processing Language · Speech interface

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1 Introduction

English teaching to foreign speakers especially students has more benefits to non-native speaking regions and more so to the learners preparing to seek academic and job opportunities in the outside world away from their homeland (Azimova & Solidjonov, 2023). China estimates that 300 million people are undertaking English learning either in school institutions or through applications which are readily available online (Gil, 2021). However, practicing the taught lessons is noted to be a major challenge to find according to Chew and Cerbin (2021). For students in schools like colleges, old classroom setups help with the teaching and due to the high number of learners to teachers which can be in the ratio of 18.32 to 1 (Gerard et al., 2024). This cannot allow the teacher to have one practice of speaking which now turns to become a student initiative to make practice on their own which equips them both in written and verbal understanding of English (Al-Jarf, 2022). Not practicing the speaking English may have the learners excel in their written exams but hinder verbal communication (Phan et al., 2022).

The benefit of artificial intelligence is being felt in all sectors and education is a beneficiary as well (Dwivedi et al., 2021). In today's era, students can polish and enhance the skills and teaching they receive in the classroom from their teachers and together they improve the quality of education (Lakkala et al., 2021). Currently, college students in non-native English-speaking countries who learn English as a second language have to more practice to perfect their speaking, and grasping as well as ensure they pass their exams through listening and repeating (Ouanhlee, 2023). English pre-recorded materials provided by the institutions due to the lack of enough personnel that can have one on one practices with the many students in a class (Yasmin et al., 2024).

Artificial Chatbots have come up to solve this issue. The advancement of machine learning language natural language processing (NLP) and speech recognition which are a result of artificial intelligence have shown great improvement in the development of chatbots (Skrebeca et al., 2021). They work by imitating and simulating the physical process of a normal conversation humans could have by being the second "person" in a conversation and conducting a dialogue with the person giving commands to it (Skantze, 2021). English as a second language could be intimidating to learners who have not learned it from their early childhood studies and this could be felt in the inadequacy to make practice with other fluent speakers (Haufiku et al., 2022). Therefore, Chatbots help bridge this gap by offering a safe space in which college learners can practice without feeling judged for the errors they make also through the application, the students can make numerous repeats and practice without the bot feeling fatigued or occupied (Kooli, 2023). A conversational bot helps the learner enjoy more learning English by being given a variety of topics which the application gives the learner depending on the information requested from the learner. This fun part of learning is very essential as the learner voluntarily makes time to practice their spoken English and within a period, they can fluently articulate their ideas and fluently keep conversation without stammering or wrongly pronunciations of words (Nguyen et al., 2021).

Looking at the English Liulishuo bot which is an English learning bot available online that anyone can access and use to learn English (Yuan, 2024). 2018 statistics show that over 70 million people have registered globally and out of that all 379 cities are in China (Wang et al., 2021). This shows the demand for learning the English language as a second language in non-native-speaking regions who are not necessarily students. In the educational sector, it is evident that chatbots promote involvement and improve performance due to their efficiency and availability to use (Kedi et al., 2024).

In this study, we developed an artificially intelligent English conversational bot which is to be used in college students who are learning English and is aimed at equipping their performance when it comes to communication as well as articulation and also proper choice of words when making a conversation. This article makes an experimental study of an Artificial intelligence conversational bot against a listen and repeated old interface for the spoken English language. We sourced the learning materials from IELTS a standardized English examination platform that covers a wide range of scenario scenes that international students could get in English university tests. Sampling 56 students from the English department in college who are learning English as a second language for the first time at the tertiary level of education. The experiment took five days and we divided the students into two groups. One group was to be involved in the experiment on fixed and free usage. To measure how effective each interface was we assessed the students' performances on how much they were able to memorize and speak vocabulary, as well as speak in their native language and have the interface translate to English so the students could understand the meaning. After the period we had stipulated for the assessment, the results showed that the learners who used the Conversational bot were more eloquent and could communicate better than the students who used the listen-repeat interface. What they confirmed was that the bot was more engaging to interact with and they were able to communicate even the hard vocabulary that they could identify through writing but could not speak them. Another important factor we captured was that the learners became more and more interested in widely learning English even outside their course bracket which means they found it more valuable to learn with.

This research lies in the development of an AI-powered conversational bot which integrates adaptive feedback mechanisms, real-time speech recognition, and natural language processing to enhance language learning. Better than the traditional chatbots, our piloted system provides personalized feedback, simulating real-life conversational scenarios, and therefore offering dynamic learning materials tailored to individual learner needs. In addition this research introduces a dual-evaluation framework under both fixed and free usage conditions, providing a comprehensive analysis of the chatbot's effectiveness compared to traditional methods. Making three contributions to the literature earlier done on how Artificial intelligence is impacting the education sector. While existing systems provide generic language training, they often lack adaptive feedback or fail to simulate real conversational dynamics. This study addresses that gap by designing and evaluating a novel English conversational bot tailored for college students learning English as a second language.

- First, we designed an English conversational bot speaking English language by

incorporating the advancement of artificial intelligent speech recognition and natural language processing.

- Secondly, the study conducts two novel empirical studies which evaluate our designed AI-enabled two-way dialogue interface against an old which listens and then repeat interface for English learning as a second foreign language speaking under both fixed and free use conditions.
- Thirdly we present a sample design which could be used as a foundation improved dialogue systems which are effective and also very engaging to the English learning for students learning it as a second language.

The remainder of this paper is structured as follows: Section 2 reviews related work in AI-driven language education. Section 3 details the design and development of the conversational bot. Section 4 describes the methodology and experimental setup. Section 5 presents and analyzes the results. Section 6 discusses the implications and limitations. Finally, Section 7 concludes with recommendations for future research.

2 Literature review

Intelligent Tutoring systems which utilize artificial intelligence techniques give personalized guidance which is solely dependent on the input given to the system by the learner who is the user (Sajja et al., 2024). Several studies have been done on the effectiveness of Intelligent Tutoring systems when compared to human tutoring. Dermevalet al. (2018) in their study on the same state that Intelligent Tutoring systems were ranked a top achiever in a study they conducted when compared to a teacher-led group or other forms of speaking guidance. The introduction of AI modules proved to bring results that moved the performance of the students from 0.29 to 1.1 standard deviations compared to other learners who were using the system in the same classroom (Fazlollahi, 2021). The first Intelligent Tutoring system developed was the German E-Tutor which was designed for adult learners learning German as a second language for non-native German speakers (Höhn & Höhn, 2019). The E-tutor was designed using Natural language processing in the process of parsing the responses learners gave to the system and it then gave specific feedback highlighting errors and gave suggestions on how to correct the errors (Bauer et al., 2023). Empirical studies showed that learners who used E-tutor made greater improvements and were more interested in correcting the errors which the system highlighted and generally their German learning was high in written and grammatical (Pandarova et al., 2019). Other Intelligent Tutoring systems are like the BANZAI and Nihongo-CALI which is discussed by Ruan (2021) in his empirical studies showed that intelligent metalinguistic feedback given by the respective system was more significant compared to other old listen and repeat that could only highlight the error but not suggest how to rectify as backed up by Zimmerman and McMeekin (2019). Therefore, the massive improvements which are seen in these systems were recommended for self-improvement when it comes to learning a second foreign language. These researches however did not use conversational bots in their Intelligent Tutoring systems for language learning therefore creating a chatbot is beneficial as it will help in improving more English

learners speaking fluent English which they learn as a second language at the college level of education.

Conversational bots allow more non controlled dialogue between the system bot and the user. The user is given control and can ask any question or start a new direction of conversation and the system automatically takes to that direction (Khatri et al., 2018). A study by Heffernan and Koedinger (2002) shows that chatbots increase learner engagement and Mousavinasab et al. (2021) affirms how much students who use both have increased their use and involvement in the system in practicing their speaking and learning of a language (Ma et al., 2014). Artificial intelligence Intelligent Tutoring systems are beneficial for giving factual information in subjects like physics and mathematics (Cano et al., 2021). In the language learning department, conversational bots and chatbots are used by a cross-board age learners and studies have shown a higher enjoyment rate compared to other learners not using the bots (Kim et al., 2019). This is because they find the bot to be more approachable than when conversing with a human being. This is the greater part that the learners who use the bots do not suffer from anxiety about not getting it right or being articulate Ishikawa (2018). The increased engagement observed with chatbots may be attributed to a short-term novelty effect (Fryer et al., 2019). Liu et al. (2022) in their research study showed that interest in English learners using chatbots was reduced after one task but their interest when practicing with human beings was intact for three tasks. English Liulishuo mobile app in China uses a chatbot with tailored tutoring which helps learners make dialogues and hence practice their understanding of English learning (Yuan, 2024). Therefore, with the high number of students in colleges taking up English learning as a second language, it is not worth that regardless of increased user involvement, results are relatively good which is good for the improvement of the learner's skills. Despite these limitations, chatbot-assisted applications remain popular and effective. In China, the Liulishuo app integrates AI-based chatbots to provide tailored English language tutoring, enabling users to engage in practical dialogue scenarios (Yuan, 2024). Given the growing number of college students learning English as a second language, the integration of conversational bots into ITS frameworks holds strong potential for improving speaking fluency and overall learner performance.

3 Methodology

3.1 Comparative experimental design

In this study, we conducted a comparative experimental study between the conversational bot and the old listen-and-repeat interface. The two separate studies under free and fixed usage scenarios aiming at differentiating the evaluation of effectiveness and engagement of the system. In the first study(fixed) the student is required to complete all tasks within the specific day given. This follows a strict manner without repeating or skipping any lesson. This evaluated how efficient the conversational bot under normal scenarios which mimics the normal classroom tradition being overseen by the tutor. In the second study (free usage) the students were allowed to access the chatbot

daily without any restriction of timing provided the tasks were done within the day given. The free usage mimics how self-paced studies are done without the monitoring of the tutor and therefore we were able to effectiveness of the bot under natural conditions. All students were asked for their consent before the start of the study and advised they are kept anonymous in the process of publishing.

3.2 Participants

We involved 60 learners from colleges studying English. Four learners who took part in the piloting phase, and then both were grouped into 28 learners who took part in the experiment. And out of which 26 and 19 from group 1 and 2 respectively finished the three-week afterward follow-up study. Their age was averagely 18.36 years ($\sigma=3.92$) and they studied in 12 different colleges in China and all of the schools offered English classes which equips them to join Western schools outside China. The learners are Chinese students who started learning English at their college level and have never been to English-speaking nations before. During the six-day commitment, the learners had to be available physically to avoid distraction from their normal learning schedules.

3.3 Tools and learning units

We implemented both systems using Python Flask JavaScript and CSS and were hosted in cloud server Mongo DB The machines we used were MacBook Pro and iMac with wired ears and a microphone. Once each learner had done the test, we used Zoom conference calls to conduct the evaluation. In six days, each study had six lessons. The first three lessons were on common communication skills which covered a wide range from knowing the learner to how communications could be done across all fields surrounding the learning environment. And the second phase of the next three days we repeated the same series of communication skills. The below figure shows the framework of the examinations done in the six days. The lessons from the 4th to 6th were redone in the first three units as shown in Fig. 1 below.

3.4 Evaluation measures

The study evaluates both the progress of the learner's English when using the conversational bot as well as getting their opinions about the interface. The first evaluation metric we did was the speaking proficiency skill when using the interface. The speak-

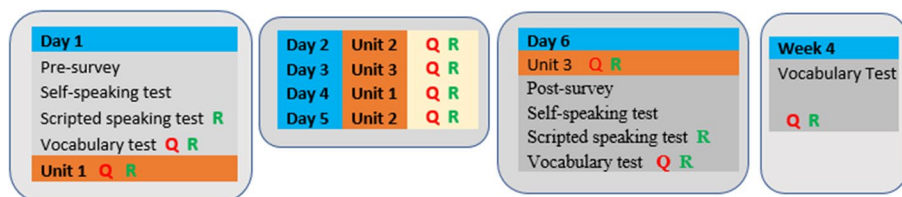


Fig. 1 Test evaluation framework

ing analysis had two sections, the self and the scripted speaking test. In the self, the learner made a conversation with an English fluent-speaking tester randomly which are in the pre-designed pool of 9 questions. Each test lasted for two minutes a maximum of five questions. In the scripted test, the learners were given Chinese scripts of the conversation they had with the tester from the pre-written script pool. Each lasted for three minutes and it needed the learner to translate the Chinese scripts into English and make conversation using the statements. We set two test materials for both the scripted and self-speaking tests. For testers, we involved three native tutors who conducted evaluations through Zoom video calls. Each tutor graded the learners participating using the rubric which we adapted from (IELTS). The tutors graded each depending on the grammatical coverage, fluency, pronunciation and accurate level using a scale of 0 to 9. Mean average of the scores was given on scale to rate a score on their speaking proficiency. The inter-rater reliability between the two graders was calculated through Krippendorff's alpha and was 0.723 which was graded the two groups using the mean average score (Gwet, 2011).

The second evaluation metric was the ability to grasp the English words and retain them. This happened after the last day of testing. To do this, we gave 36 Chinese words and the participants needed to translate them into English. The third metric was to analyze the anxiety levels of the learners using the SLSAS that builds on Foreign Language Classroom Anxiety Scale. Studies by Kasap (2021) shows that foreign language brings anxiety to learners especially those who have never learnt it at all. and using this we evaluated the anxiety of the participant during the test and after (Andrade et al., 2019).

After completing the period of using the system, we incorporated the User Engagement Scale Short Form (Andrade et al., 2019) which is measures software engagement and uses 12 questions can be answered within 15 min. Prior to our studies, we surveyed to gauge participants' attitudes toward oral English studies. Our findings revealed that the majority of participants were driven by external factors. These factors include opportunities abroad English was mandatory for them, and high-stakes exams IELTS (89.3%). The driving factors are opposed to a voluntary personal interest that enhances one's self-improvement (10.4%). Out of the basic language skills including listening, speaking, writing, and reading skills, speaking was rated as the most important (46.4%), useful (60.7%), and difficult to learn (48.2%). There were some challenges that the participants faced when speaking English. For instance, some (21.4%) are self-conscious about how they pronounce English terms and words, others (14.3%) fear making mistakes before others, and other participants 919.6%) find it difficult to find people whom they can practice the language with. These results highlight the importance of creating an inclusive learning platform helping students improve spoken and written English by engaging them effectively.

3.5 Procedure

The study conducted two in subjects' analysis, having each subject follow a similar flow of study as illustrated in Fig. 3. Each of them took 6 days consecutively. On the first day, users were given a pre-survey that contained an anxiety measure for a foreign language. Then a speaking test and lastly vocabulary testing. The users were

then randomly selected to and allocated the interface for use, either an old-style or a Conversational bot for the period of evaluation. On the study's last day, post-survey was conducted using the same tester whereby participants were subjected to the same anxiety measure as on the first day, including the vocabulary and speaking tests and a similar engagement style. We first had to predetermine the number of practices by conducting a fixed use between-subjects study. The next study on free usage and the users freely used the system. Our goal was to determine the level of engagement people experienced with both systems in the first study and to observe how practising the same number of sentences impacted people's English-speaking ability in the second study.

3.6 Between-subjects study

In the first study (Fixed usage), participants were instructed to follow the guidelines of the learning systems precisely. With the Conversational bot, learners completed each lesson only once. During each lesson, they listened to the questioner's audio prompts automatically generated by the Conversational bot. They responded to the hints provided by the system, recording their answers for feedback. Once they received feedback, they clicked "reference answer" icon/button. This produced an answer that they read, and listened to the audio at one instance. Old system interface, learners had an advantage as they were able to listen to all audio recordings in the order presented. They were instructed not to repeat the questioner's content after listening to the audio, but to repeat the responder's content once after listening to the audio. In the second study 2 (Voluntary usage), participants engaged with the systems at their discretion to improve their oral English skills over six days. Each day, learners were granted access to a unit, as illustrated in Fig. 3 above. Usage time was not restricted, and individuals were encouraged to explore all of the features offered within the unit as frequently as they desired.

3.7 System design

This study gives a design and implementation of an English conversational bot which leverages recent Artificial intelligence advancement. The bot is designed for college students in non-native-speaking regions who take up English in preparation for joining higher schooling in English-spoken regions. To evaluate the feasibility of the system, this study uses an old listen-and-repeat interface. This old system helps draw the baseline for our design.

The conversational bot had four distinct parts which interplayed together for the effectiveness of the system. We had the learning materials, the interface platform, the speech recognition system, and the adaptive feedback system. All these were designed to work together for the effectiveness of the bot.

3.8 Learning materials

From the International English Language Testing System (IELTS) website, we used four conversations from the listening section. The reason for using this conversation

is because IELTS exams are used by universities to evaluate the English competency of the learners joining the institutions. Therefore, using these conversations brings the learners to be acquainted with the level of results required and which they will sit for during their time of exam. The conversation picked lasts for 15–20 min which is adequate for a lesson. The level of difficulty too is good. The question ranges from easy to difficult and this is good for balancing and knowing the level at which each learner is best comfortable. To ensure there is fairness and clarity in evaluating, the test is made of two roles the “Questioner” and the “responder”. As per the conversational bot we designed the first to initiate the conversation is the questioner and the later to respond is the responder. Both the learner and the system can take either lead, however, at the start of the use, the system plays the questioner as it takes the lead in the conversation.

Both the old listen and repeat interface which we use as our baseline and our designed AI English conversational bot are loaded with the same four conversations which act as our lessons for tutoring. The lessons are universal and they relate to English learning but around learning in college.

3.9 English conversational bot interface

We implemented the conversational bot system using Python Flask as a web application with two independent panes and named it Xuéxí which leverages a human persona effect [30]. The first right side pane shows all the learning materials extracted from the IELTS website which mainly are the sentence structures and the vocabulary and the left side shows the voice chatbot recorder part which the student could make conversation using it. The left side is where the learner interacts more with the bot and has three sections as shown in Fig. 2 below.

The interface has eight parts as labelled; (1) Conversation practice side. (2) Learning material side. (3) Words, with “play-audio” as well as “translation” icons. (4) Learner’s words using conversation bot (5) Response/feedback, and the “play-audio” button. (6) “Hint” bar. (7) Operation panel showing the first phase, allowing the learner to speak out loud to record their response. (8) Comes up after the learner’s voice is recorded and changed to text. Learner can select “mic” button through a click to proceed with recording of their words or write on the text part.

- The far top part has conversation bubbles between Xuéxí and the learner. This includes the “audio-play” icon/button which replays recorded English words and sentences audio, “translation” icon/button, that translates English words and sentences to respective native languages for ease of understanding the meaning and the chatbot utterances. The learner’s dialogue bubbles contain the learner’s answers, Xuéxí’s feedback, the “reference answer” that allows the learner to see the reference answer from the main sentences conversation and lastly the “audio play” button that reads the reference answer to the student.
- The middle section is the “hint” section. This part has the suggestion and prompts in the student’s native language.
- The bottom section is the operation panel. This is where the student does the task of recording, editing and sending responses to Xuéxí’s prompt.

FINISH LESSON

Hello (你好) 3

How are you? (你好吗?)

Hello too. 4

I am fine, thank you.

How can we help you today?

5 (我想知道如何注册我的课程，还想知道如何找到一个好的宿舍。)

I would like to know how to register my course and also know how to find a good hostel

4

Okay, Please register your course online. For the hostel, once you register, you will also find on the portal information.

Kindly choose, around the school for ease of transport.

6 请说：“交通便利”是什么意思

7

8 That means easy ways of getting to school

Lesson One 2

Vocabulary	
Hello	你好
How are you?	你好吗?
My name is	我的名字是
I live in	我住在
Welcome	欢迎
University	大学
Report	报告
Apply for scholarship	申请奖学金
Passport	护照
Education level	教育水平
Hostel	宿舍
Restaurant	餐厅
Driver's license	驾驶证
Course	课程

Sentence structure	
It is my pleasure to meet you	这是我的荣幸认识你
Kindly reach out in case of any issue	如有任何问题，请随时联系
Please register your course online.	请在网上注册您的课程

Fig. 2 The English conversational interface

Xuéxí plays the main role of being a female tutor as well as the practice companion to the student. All audio content which includes instructions and the content of the dialogue lessons was recorded by a female native English speaker. Which makes it in line with our conversational bot-chosen female tutor named Xuéxí. The reason for choosing the pre-recorded audios was because we need a more natural mimic of a human tutor which gives the student a relaxed space when practicing which is the main aim of our conversational bot of mimicking human beings. Secondly, the prerecorded content helps the students make comparisons of what they hear to how they make pronunciations as well. Which also helps them improve their pronunciation skills. The conversational bot starts the communication by first making an introduction to how the student can navigate through the system. This gives the student the direction on how to use the system, and also what to expect of the system. It also gives an overview of the course content. Once that is done the lesson begins and the bot initiates the talks and gives hints to the student about their chosen native language., Based on the hint, the students respond to the communication given by the bot through the hints given in the “hint bar”. Once that is done conversational bot automatically plays the audio. Once the student hears the audio, they give a voice response which the system captures and transcribes into text format. The student can

replay the text severally until they are sure they correctly heard and then give the response. Once the student's audio is transcribed by the system the student can edit the text field to amend what they responded. They can also append new statements by making a new recording and discarding the previous record. Once the student is confident of their response, they approve the message to be sent to the system and immediately the conversational bot gives feedback on the response the student gave. The system gives the learner the ability to view reference answers from the original conversation by reading the text and also listening to the audio recording of the specific reference answer's vocabulary and sentence structures in the learning material pane.

3.10 Speech recognition

For speech recognition we needed both the software and hardware and for the software part, we used Amazon Transcribe AI-powered for speech recognition and transcription. For the hardware, we used a wired microphone placed close to the student's mouth for audibility. Once these were set up, we sampled 200 spoken words by the students to check on how accurate the speech recognition would be and from the results we had over 95% accurate transcription. We noted that the 5% which were inaccurate were either:

- near-homophones and indistinguishable liaisons words which are words that are pronounced almost the same like “farther/further”, “cellar/seller”, “cymbal/symbol”, “working/walk-in”.
- complex long words which are words that are long and difficult to pronounce for example “phenomenon”, “indistinguishable”.

3.11 Adaptive feedback system

We implemented a personalized guidance system in a conversational bot which compares the learner's transcribed response with the correct response and gives a scoring rate to the learner's response. The rating score is based on two criteria. The first is the length score. This criterion engages the student more gives penalty for short phrased answers and encourages the student to speak more. We computed the length score as the ratio of the length of the student's response to the length of the correct answer and we capped it at 1. The second criterion is the semantic similarity between the responses the student gave and the correct responses which was computed using the SIF algorithm as cosine similarity.

The students who used the pilot phase of the conversational bot without feedback did a total of ($n = 264$). They rated the quality feedback they expected from the system based on their responses against textbook responses. The ratings were on three levels. The highest being 3 which meant “excellent”, average 2 with “good” and the lowest 1 with “okay”.

The bot we created was meant to help students improve and give a good and easy environment to learn and therefore the use of negative words was excluded from our system. Using the pilot students' responses and labels we took the length score and

similarity score of the responses as features and fitted a logistic regression model using 214 pairs of pilot students' answers and labels. To validate the scoring algorithm, we tested this using the 50 extra pairs of responses and labels. With that, got mean absolute error (MAE) of 0.52 which is about 50% level on the training set and 0.55 on the test set. This validates the new system to grade the students' responses on the three levels, "Okay", "good" and "excellent". Each level has been designed to have 10 positive feedback statements which accompany the response. This is based on the grading level which the system, puts the students based on their responses from a corresponding pool. Each feedback remark statement does not tell from which pool it was pulled from to the student.

3.12 Old system listens and repeat interface

In this study we created a non-AI style which has an interface that listens and repeats based on English learning software (Lanren-English) used for speaking shown below in Fig. 3.

The "listen and repeat" structure interface has the right and left panes. The right pane has the learning materials and the left displays the dialogue section. The student can listen to a recorded audio conversation and also read transcribed texts of the conversation. The system only allows the student to replay as many times as they want but they cannot input anything to the system.

FINISH LESSON 1			Lesson One 2	
			Vocabulary	
Hello	(Translation)	(你好)	Hello	你好
Hello too.	(Translation)	(你也好)	How are you?	你好吗?
How are you?	(Translation)	(你好吗?)	My name is	我的名字是
I am fine, thank you.	(Translation)	(我很好, 谢谢你)	I live in	我住在
Okay,		好的	Welcome	欢迎
How can we help you today?	(Translation)	今天我们能为您提供什么帮助?	University	大学
I would like to know how to register my course	(Translation)	(我想知道如何注册我的课程)	Report	报告
Please register your course online.	(Translation)	请在网上注册您的课程	Apply for scholarship	申请奖学金
Okay, how about hostel	(Translation)	好的, 青年旅舍怎么样	Passport	护照
For the hostel, once you register, you will also find on the portal information.	(Translation)	对于宿舍, 一旦您注册, 您也会在门户网站上找到信息	Education level	教育水平
			Hostel	宿舍
			Restaurant	餐厅
			Driver's license	驾驶证
			Course	课程
			Sentence structure	
			It is my pleasure to meet you	这是我的荣幸认识你
			Kindly reach out in case of any issue	如有任何问题, 请随时联系
			Please register your course online.	请在网上注册您的课程

Fig. 3 Old system listen and repeat interface

3.13 Database

To ensure that the system worked well, we stored all actions needed to use both systems in a MongoDB database [33]. This included button clicks, recordings, timestamps and all user actions. The reason we chose MongoDB is because it is a Full cloud-based developer data platform and is simple to install which is what we aimed at.

4 Results

Before running all the t-tests we conducted the Shapiro-Wilk normality tests. In both studies, we evaluated three main metrics: Time spent on the application, engagement and anxiety change, Vocabulary Test and lastly Speaking Test.

4.1 Time spent on the application

The findings show that in the first study (*Fixed*), controlled use condition, learners spent average total of 45.07 ($\sigma=11.14$) and 103.74 ($\sigma=29.63$) minutes on the app respectively on old and conversational bot as illustrated in (Fig. 4) and Table 1.

In the second study (*voluntary*) we noted that conversational bot learners used the system twice the period than old system learners significantly ($t_{26}=6.1$, $p<0.0001$ and Cohen's $d=2.30$). As shown in Fig. 4; Table 1, the six-day learning period was consistently higher for Conversational bot learners. Additionally, the Conversational bot was rated to engage learners than the old interface system ($t_{26}=2.8$, $p<0.05$ and Cohen's $d=1.04$). It averagely rated at 4.36 ($\sigma=0.43$) while the old system rated 3.87 ($\sigma=0.51$).

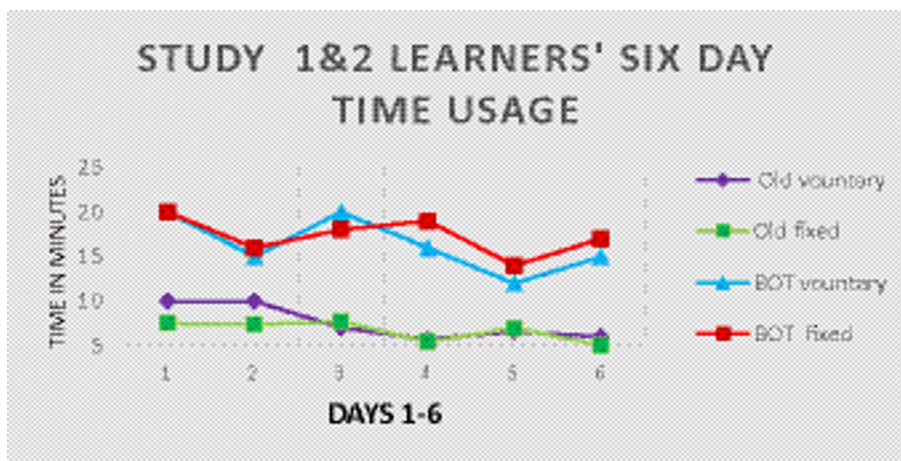
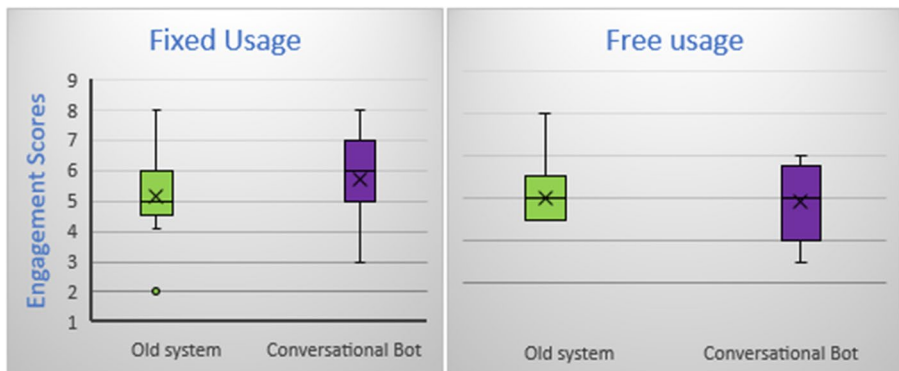


Fig. 4 Six-day learners' time usage

Table 1 Time spent (free and fixed usage)

Fixed study (Study 1)					Fixed study (Study 2)				
Old system before test			New system before test		Old system after test			New system after test	
Standard deviation		10.75		26.53	Standard deviation		8.70		32.24
Average		46.14		104.02	Average		36.96		113.45
Students	System	Time Spent	System	Time Spent	Students	System	Time Spent	System	Time Spent
1	OLD	58.33	NEW	76.82	1	OLD	23.82	NEW	134.24
2	OLD	43.61	NEW	129.02	2	OLD	41.84	NEW	117.93
3	OLD	55.63	NEW	140.03	3	OLD	29.26	NEW	98.43
4	OLD	23.19	NEW	81.38	4	OLD	46.35	NEW	103.42
5	OLD	43.13	NEW	130.06	5	OLD	30.85	NEW	113.9
6	OLD	27.31	NEW	161.9	6	OLD	32.36	NEW	130.21
7	OLD	46.89	NEW	83.37	7	OLD	37.5	NEW	148.77
8	OLD	44.39	NEW	79.34	8	OLD	50.13	NEW	186.3
9	OLD	43.87	NEW	81.53	9	OLD	27.38	NEW	124.24
10	OLD	44.94	NEW	115.81	10	OLD	43.97	NEW	80.3
11	OLD	43.66	NEW	106.09	11	OLD	49.78	NEW	120
12	OLD	61.85	NEW	97.86	12	OLD	26.38	NEW	103.54
13	OLD	48.39	NEW	73.97	13	OLD	33.51	NEW	80.61
14	OLD	60.82	NEW	99.08	14	OLD	44.34	NEW	46.35

**Fig. 5** Study 1&2 self-reported learner engagement score

4.2 Engagement

In the first study (*fixed*) the results show that those who used the Conversational bot spent approximately 2.6 times longer than those on the old system thus statistically significant difference having large effect size lot (Cohen's d [6] = 2.90), as proven by the two-sample t-test ($t_{26} = 7.6$ and $p < 0.0001$). After the end of the learning period, user engagement scores were evaluated and self-reports were displayed as in Fig. 5 below. And Table 2.

Table 2 Study engagement score ratings

Fixed (study 1)				Voluntary (study 2)			
System	Engage- ment Score	System	Engage- ment Score	System	Engage- ment Score	System	En- gage- ment Score
OLD	4.07	BOT	5.36	OLD	3.64	BOT	4.16
OLD	4.05	BOT	3.56	OLD	4.31	BOT	4.83
OLD	4.16	BOT	4.66	OLD	3.98	BOT	4.05
OLD	4.42	BOT	3.67	OLD	3.23	BOT	4.38
OLD	4.41	BOT	4.39	OLD	3.96	BOT	5.28
OLD	4.13	BOT	3.9	OLD	3.42	BOT	4.63
OLD	4.13	BOT	4.65	OLD	3.95	BOT	4.44
OLD	5.83	BOT	4.05	OLD	3.9	BOT	4.73
OLD	4.16	BOT	4.45	OLD	3.29	BOT	4.02
OLD	4.45	BOT	4.08	OLD	3.17	BOT	4.31
OLD	4.2	BOT	4.16	OLD	3.39	BOT	4.73
OLD	3.48	BOT	4.08	OLD	4.13	BOT	4.08
OLD	3.91	BOT	4.58	OLD	3.75	BOT	5.12
OLD	3.84	BOT	4.53	OLD	3.71	BOT	4.26
OLD	3.49	BOT	4.22	OLD	3.62	BOT	4.36

T-test ($t_{26} = 7.6$) $p < 0.0001$ Cohen's d [6] = 2.89

The report shows that the Conversational bot learners had a higher rating than the old system ($4.18 > 3.95$). This difference was insignificant according to the two-sample t-test that showed that $t_{26} = 1.1$, $p > 0.5$. In the second study (*voluntary*) the findings show that learners of the first study, learners when practicing with conversational bot spent an average of 97.0 ($\sigma = 19.8$) minutes learning, while learners of the old interface spent an average of 46.5 ($\sigma = 23.9$) minutes.

4.3 Anxiety

As for the anxiety level changes SLSAS was used to evaluate the change whereby both systems. In the first study (*fixed*), the conversational bot and the old systems varied narrowly (both $ps > 0.05$), as per the one-sample test as both decreased after the 6 days. As for the Conversational bot learners, the anxiety decreased by 0.03 ($\sigma = 0.34$), while that of the old system decreased by 0.12 ($\sigma = 0.59$). Additionally, the tests demonstrate that the two systems had done major significant difference in terms of affecting the levels of anxiety ($t_{26} = 0.5$, $p > 0.05$). In the second study (*free*) we noted that despite the slight increase in speaking English, anxiety changes in both groups, neither had a significant change ($ps > 0.05$) and there was a small similarity between both old and new systems ($t_{26} = 0.4$, $p > 0.05$). Overall, these findings suggest that a Conversational bot is a more effective and engaging tool for casual language learning as compared to the old system as shown in Fig. 6; Tables 3 and 4 below.

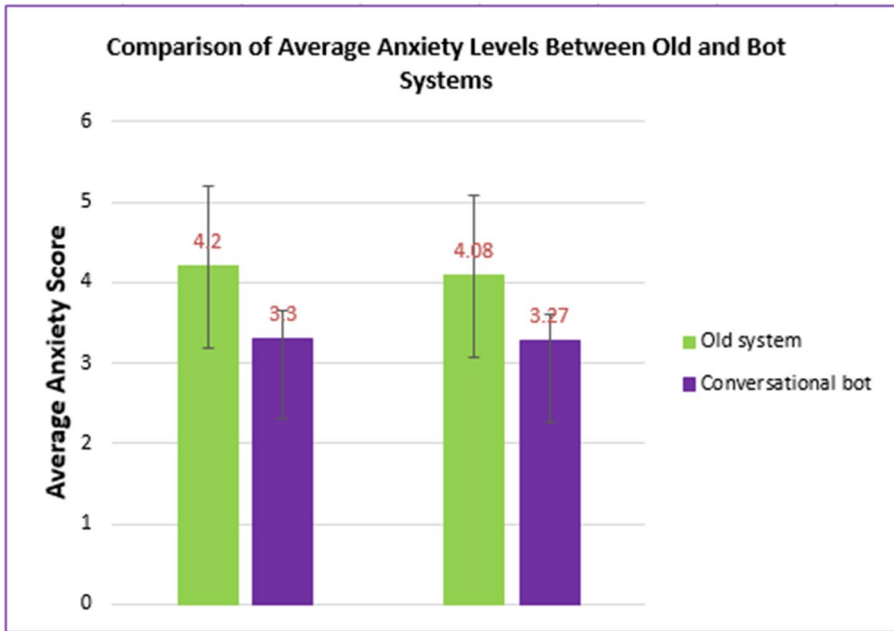


Fig. 6 Anxiety comparison graph

Table 3 BOT's anxiety comparison scores

Old system before test					Old system after test				
Item	Avg Mean	Std Dev	Cron-bach's alpha	Items no	Item	Avg Mean	Std Dev	Cron-bach's alpha	Items no
Cognitive anxiety	4.3	0.74	0.81	3	Cognitive anxiety	4.2	0.79	0.81	3
Somatic anxiety	4.2	0.79	0.90	3	Somatic anxiety	4.2	0.78	0.90	3
Behavioral anxiety	4.2	0.79	0.86	3	Behavioral anxiety	4.2	0.77	0.86	3

Table 4 BOT's anxiety comparison scores

BOT system before test					BOT system before test				
Item	Avg Mean	Std Dev	Cron-bach's alpha	Items no	Item	Avg Mean	Std Dev	Cron-bach's alpha	Items no
Cognitive anxiety	3.1	0.78	0.81	3	Cognitive anxiety	3.3	0.80	0.81	3
Somatic anxiety	3.1	0.77	0.90	3	Somatic anxiety	3.1	0.80	0.90	3
Behavioral anxiety	3.3	0.79	0.86	3	Behavioral anxiety	3.2	0.78	0.86	3

4.4 Vocabulary test

To account for differences in learner's interactions with resources in both the responder's and questioner's roles we analysed vocabulary test results separately for each role and adjusted them using Bonferroni's procedure. After the 6-day learning

period, the test showed that the old system learners had acquired 4.5 new words ($\sigma=2.7$) from questioner's side ($p<0.001$ and Cohen's $d=2.36$) and 5.5 new words ($\sigma=2.7$) from responder's part ($p<0.0001$, Cohen's $d=2.14$). This accounted for the initial progress from the initial state to the end of the six days such that the learners had learned 62.8% ($\sigma=22.6\%$) of the new words from questioner's role and 81.2% ($\sigma=21.0\%$) from responder's role. As for the Conversational bot learners, they had learned 3.9 new words ($\sigma=2.4$) on the questioner's side ($p<0.001$ and Cohen's $d=2.03$) and 8.7 new words ($\sigma=2.9$) on the responder's side ($p<0.0001$ and Cohen's $d=4.32$). This accounted for 44.1% ($\sigma=21.9\%$) of new words on the questioner's side and 88.1% ($\sigma=16.9\%$) of new words on the responder's side. When the two are compared according to the two-sample tests, it is evident that the Conversational bot learners learned more words than the old system learners, such that the results were ($t_{26}=3.4$ and $p<0.005$ and Cohen's $d=1.27$) on the responder's role and ($t_{26}=0.7$ and $p>0.05$) on the questionnaires' role. Figure 6 also shows results from a three-week delayed vocabulary test, which indicated that the learner's vocabulary had improved significantly in both systems in both roles ($p<0.001$ for old system on the questioner's role, $p<0.01$ for Conversational bot on the questioner's role and $p<0.001$ for the old system on the responder's role, $p<0.0001$ for the Conversational bot on the responder's role, though they did decrease over time. Conclusively, both systems were not significantly different from each other for either role ($ps > 0.05$) as in Figure 7 below.

According to the questionnaire results, learners using the old system learned an average of 5.6 ($\sigma=2.9$) new words, representing 71.7% ($\sigma=22.8\%$) words from pre-test, and this improvement was significant ($p<0.0001$). Learners using the Conversational Bot acquired an average of 4.1 ($\sigma=2.7$) new words, or 59.1% ($\sigma=28.4\%$) of previously unknown words, and their improvement was also significant ($p<0.001$). On the responder's side, learners using the old system memorized 5.1 ($\sigma=3.6$) addi-

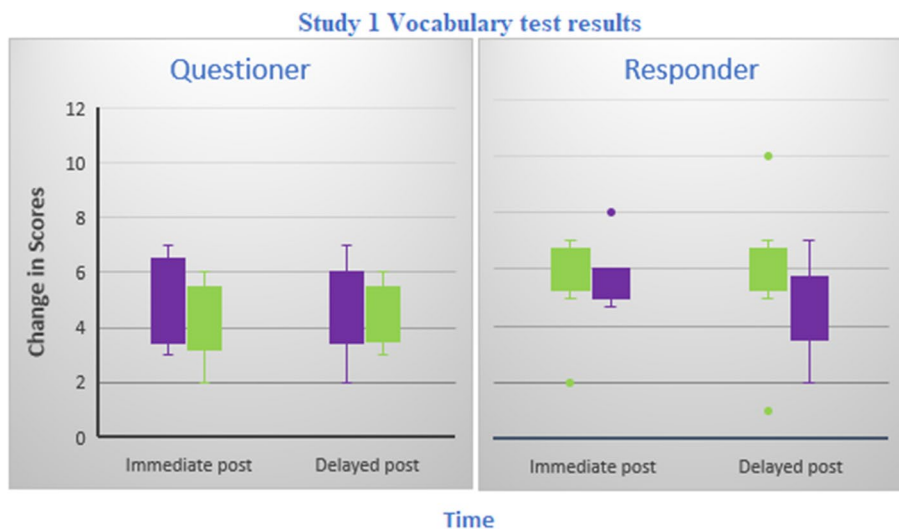


Fig. 7 Vocabulary scores

tional words ($p<0.001$), marking a 74.9% ($\sigma=21.5\%$) improvement compared to their initial vocabulary test. In comparison, Conversational Bot learners memorized 7.0 ($\sigma=3.0$) new words ($p<0.0001$), equivalent to a 93.8% ($\sigma=11.4\%$) improvement from their initial test performance. No significant difference in vocabulary improvement was observed between the two systems for either the questioner's side ($t_{26}=1.6$, $p>0.05$) or the responder's side ($t_{26}=1.7$, $p>0.05$).

During three-week delayed vocabulary test, learners using the old system retained an average of 2.4 ($\sigma=6.7$) words from questioner's side ($p<0.0001$) and 2.1 ($\sigma=6.8$) on the responder's side ($p<0.01$). Learners using the Conversational Bot retained an average of 2.9 ($\sigma=1.7$) words ($p<0.01$) from questioner's side and 5.5 ($\sigma=3.0$) on the responder's side ($p<0.01$). There was no significant difference between the two systems on both delayed vocabulary test results for either the questioner's side ($t_{17}=0.2$, $p>0.05$) or the responder's side ($t_{17}=1.9$, $p>0.05$), as shown in Fig. 8, which illustrates learners' vocabulary score improvements.

4.5 Speaking test

Using the 10 one-sample t-tests over a period of 10 days to evaluate changes in sub-scores for grammar, fluency, pronunciation, lexicon, and overall average scores for both interfaces. The results from Bonferroni correction mean to do many comparisons indicated that the Conversational Bot resulted in a significant improvement in lexicon sub-scores on the scripted speaking test ($p<0.05$) having (Cohen's $d=0.94$) which is high. Conversely, old interface showed great improvements across every sub-scores: fluency which was at (Cohen's $d=0.48$) and $p<0.05$, grammar (Cohen's $d=0.68$ and $p<0.0001$), lexicon ($p<0.0001$, Cohen's $d=0.87$), and pronunciation ($p<0.05$, Cohen's $d=0.43$), as well as the overall average speaking test scores ($p<0.0001$, Cohen's $d=0.68$). On the free-form use speaking test, there was no significant improvements in any sub-scores or overall scores in both systems, as depicted in Fig. 9; Table 5. Furthermore, two-sample t-tests revealed no significant differences between the sys-

Study 1 Vocabulary test results

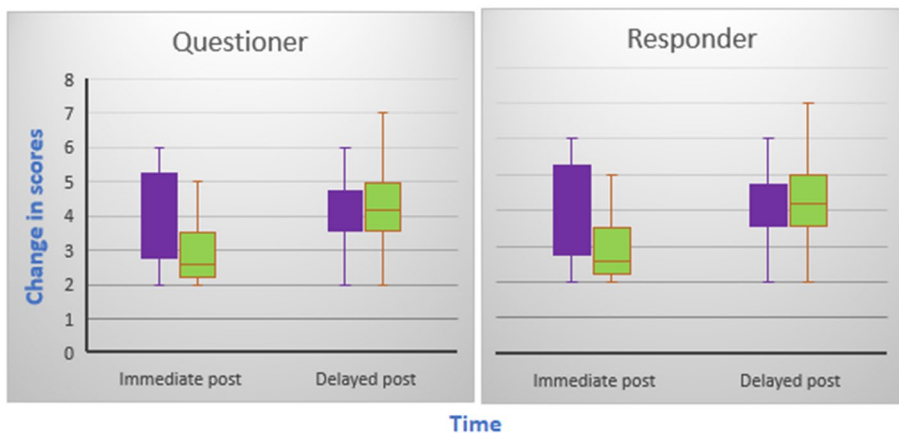


Fig. 8 Vocabulary test results

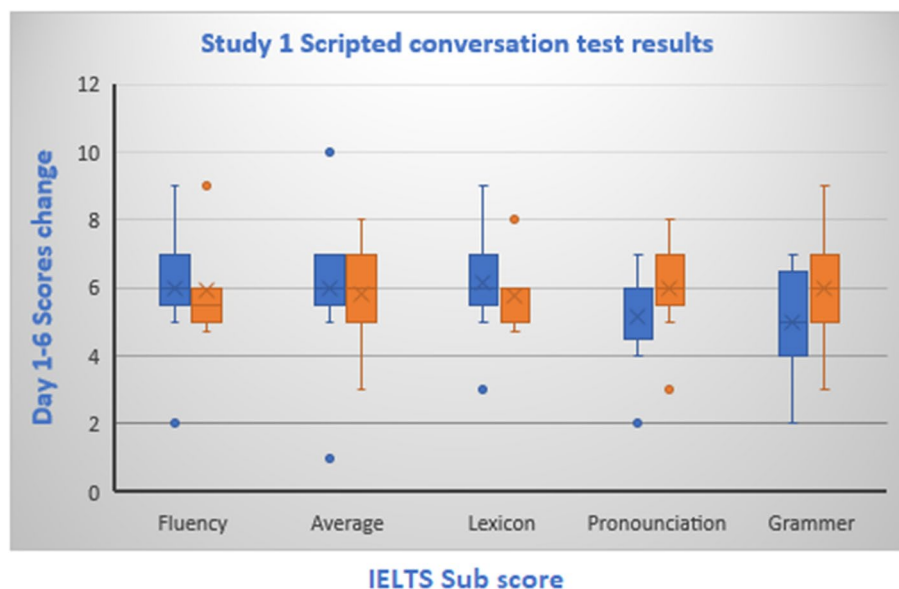


Fig. 9 Conversation test results

Table 5 Study 1 Improvements in script-based and free-form conversation test scores from pre-test to post-test

Study 1 Improvement results			
Sub-score	Interface	Scripted	Free-Form
Pronunciation	Old system	0.393*	−0.214
Pronunciation	Conversational Bot	0.179	0.214
Grammar	Old system	0.643***	0.214
Grammar	Conversational Bot	0.821	0.25
Fluency	Old system	0.500*	−0.036
Fluency	conversational Bot	0.607	0.214
Lexicon	Old system	1.000*	0.036
Lexicon	conversational Bot	0.964*	0.107
Average	Old system	0.634***	0.000
Average	conversational Bot	0.643	0.196

tems regarding improvements in the four speaking test sub-scores or overall average scores for both scripted and free-form speaking tests ($ps > 0.05$).

The one-sample t-tests showed that the Conversational Bot led to notable language improvement when gauging speaking test on the scripted test ($p < 0.05$ which is Cohen's $d = 0.64$), with significant gains specifically in fluency ($p < 0.01$, Cohen's $d = 0.84$) and lexicon sub-scores ($p < 0.05$, Cohen's $d = 0.83$). Both systems failed to show significant changes in any sub-scores or overall results on the free-form speaking test ($ps > 0.05$). The two-sample t-tests further indicated that the Conversational Bot provided significantly larger gains in the fluency sub-score ($t_{26} = 3.1$, $p < 0.05$, Cohen's $d = 1.18$) for the scripted test, though no significant differences emerged for other sub-scores or overall scores ($ps > 0.05$). Neither additionally, when fluency improvements were normalized by the period used thus statistically non-significant

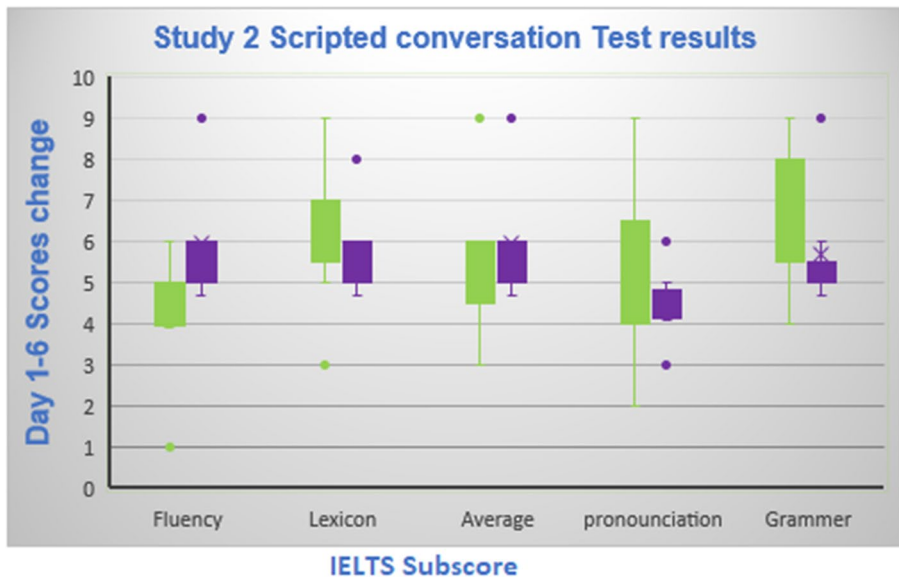


Fig. 10 Scripted conversation results

Table 6 Scripted conversation results

Study 2 Improvement results			
Sub-score	Interface	Scripted	Free-Form
Pronunciation	Old system	−0.143	−0.143
Pronunciation	Conversational Bot	0.143	0.107
Grammar	Old system	0.000	0.036
Grammar	Conversational Bot	0.429	−0.036
Fluency	Old system	−0.036	−0.214
Fluency	conversational Bot	0.697**	0.000
Lexicon	Old system	0.107	0.071
Lexicon	conversational Bot	0.786*	−0.107
Average	Old system	−0.018	−0.063
Average	conversational Bot	0.509*	−0.009

($p > 0.05$). system showed any advantage in improving scores on the free-form speaking test ($p > 0.05$), represented in Fig. 10; Table 6.

5 Discussions

This research provides valuable insights, highlights experimental constraints, and offers design recommendations for developing advanced conversational interfaces aimed at second language learning. The findings reveal that learners who voluntarily used the system were more engaged, leading to fewer restrictions and allowing for greater interaction with the bot compared to older interfaces. As noted in prior studies on educational chatbots, English conversational bots encourage higher engagement

levels (Goda et al., 2014). However, learners did not find the bot more engaging under fixed usage conditions. This difference could be attributed to the familiarity of older listen-and-repeat systems, which are often part of traditional classroom settings, where their use is mandatory. When learners engage with these older systems voluntarily, they may still associate them with unpleasant classroom experiences, unlike the fresh appeal of chatbot-based learning. Learners using the conversational bot in an unrestricted setting significantly improved their performance in scripted conversation tests, including fluency, vocabulary, and overall scores, while those using the older system showed no such improvement. These results suggest that incorporating chatbot features may enhance English learning systems. This limitation may stem from the short learning period provided. It is well understood that mastering a second language is a complex process requiring time (Ortega and Iberri-Shea, 2005). Thus, future research should focus on long-term studies to assess the effectiveness of conversational bots in language learning.

The feature of the Conversational Bot most commonly cited as “useful” by learners ($n=11$) was the real-time adaptive feedback provided during conversations. Offering immediate and positive feedback is essential for boosting learners’ confidence and encouraging them to engage more in speaking practice. Learners suggested that future enhancements should include more comprehensive feedback, focusing on areas such as word choice, grammar, and pronunciation. The use of high-quality pre-recorded audio for the lessons proved particularly beneficial for learners in terms of self-correcting their tone, speed, and pronunciation ($n=10$). Recording and replaying their own responses, followed by comparison with pre-recorded examples, allowed learners to reflect on their performance and make faster progress. In addition to structured practice, a few learners ($n=3$) noted that engaging in random conversations and casual chit-chat significantly enhanced their long-term engagement and speaking skills. They also suggested that verbal hints, rather than text-based ones, could make practice more effective by keeping the conversation more natural. Incorporating these design improvements and conducting long-term studies to better assess students’ speaking ability progression is essential for future development.

6 Conclusions

Inspired by the success of chatbots in other educational areas, we developed Conversational Bot, an interactive tool designed to help learners learn foreign language through conversation. In a comparison with the old system which only provides a listen and repeat interface, the chatbot proved to be more engaging. It also contributed to improvements in vocabulary acquisition, fluency, and comprehension. The findings from this study highlight the potential of AI-powered conversational bots to transform language learning environments by offering personalized, adaptive feedback and promoting active learner participation. Unlike traditional methods, the conversational bot fosters an immersive learning experience, where students feel more motivated to practice speaking without fear of judgment. The bot’s ability to provide real-time feedback, tailored to individual learning needs, encourages continuous improvement and helps learners build confidence in their language skills.

Moreover, the significant increase in voluntary usage suggests that learners find the chatbot not only effective but also enjoyable to interact with, making it a sustainable tool for long-term language development. The integration of speech recognition, natural language processing, and adaptive feedback mechanisms within the chatbot further enhances its capacity to simulate real-life conversational scenarios, preparing students more effectively for academic and professional communication in English.

In conclusion, this study demonstrates that conversational bots are not just supplementary tools but can serve as core components in language education, especially in environments with limited access to native speakers or language instructors. The positive outcomes observed in vocabulary growth, fluency, and learner engagement underscore the chatbot's value as a powerful educational technology for second language acquisition.

7 Limitation and future recommendation

This study involved a relatively small sample of 56 Chinese college students from a specific geographical region. As a result, the findings may not be fully generalizable to learners from other cultural, linguistic, or educational backgrounds. The limited demographic and cultural diversity restricts the external validity and the broader applicability of the results to international or non-college populations. Furthermore, the short-term nature of the study may not capture long-term engagement, retention, or fluency development associated with the use of conversational bots. Variations in learner motivation, prior exposure to technology, and individual learning preferences were not controlled and may have influenced the outcomes.

For future research, we recommend Drawing from qualitative feedback, the study proposed the following recommendations to create more effective and engaging intelligent conversational interfaces for English language learning. Thus by conducting longitudinal studies with more diverse and larger participant pools across multiple educational institutions and cultural contexts. Secondly examining how learner characteristics like anxiety levels, language proficiency, tech familiarity which influence the effectiveness of AI-driven tutoring systems. Third Investigating the integration of conversational bots with human feedback to understand the complementary benefits of blended learning. Lastly exploring the impact of sustained use over time and how novelty effects wear off, particularly in real classroom settings.

Data availability Data used are within this manuscript.

Declarations

Competing interests The authors have no competing interest to declare.

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